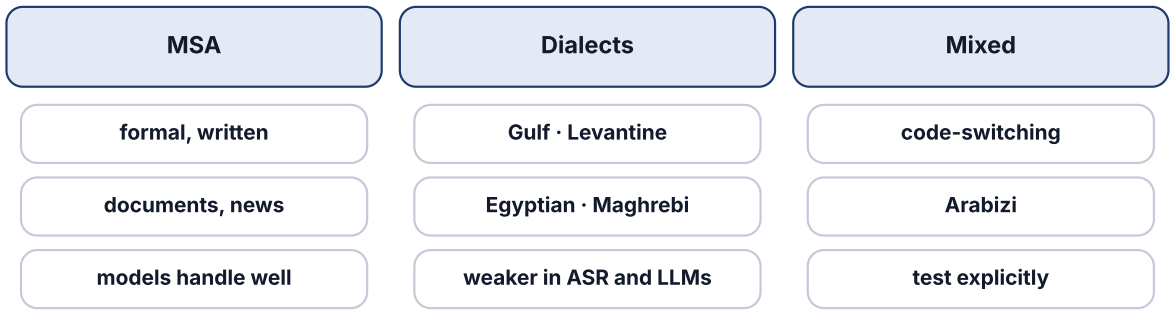
Multimodal, Voice & Arabic AI

1. The Arabic language landscape

In the Gulf, 'does it work in Arabic?' is often the deciding question.

- **MSA** (Modern Standard Arabic): formal, written, news and documents
- **Dialects**: Gulf, Levantine, Egyptian, Maghrebi: how people actually speak and type
- **Code-switching**: Arabic and English mixed in one sentence
- **Arabizi**: Arabic in Latin letters and numbers ('7abibi')

✎ DIAGRAM



Test the variety your users actually write and speak.

2. Technical issues with Arabic

- **Token tax**: ~1.5 to 2.5× English tokens for the same meaning → more cost and latency, less context
- **Normalisation** before keyword search: alef forms (ا أ إ آ), ta marbuta, diacritics, elongation
- Right-to-left and mixed-direction display
- **Speech recognition** is noticeably weaker on dialects
- Retrieval: a **multilingual embedding model**, cross-lingual testing, normalised text for BM25

3. Evaluating Arabic

- Golden sets **per dialect**, labelled by native speakers
- LLM judges can favour MSA or English phrasing, so calibrate them on Arabic
- Report quality by language and dialect, not one overall score
- Arabic-centric models exist (e.g. **Jais**, **ALLaM**): test them alongside global models on your data
- Residency: UAE and Saudi **PDPL** often require in-region processing

4. Documents and vision

- **OCR**: turns images of text into text. Cheap and fast, loses layout, struggles with tables and handwriting
- **Document AI** (Textract, Azure Document Intelligence, Google Document AI): keeps structure, tables and key-value fields
- **Vision-language models (VLMs)**: multimodal LLMs that read images directly. Flexible on messy layouts, cost more per page
- Fixed high-volume forms → document AI; varied messy documents → VLM into a schema; search → parse with structure, then chunk
- Extract into a **strict schema**, return **confidence per field**, route low-confidence fields to a human, link every value to its page, measure **field-level accuracy**

✎ DIAGRAM



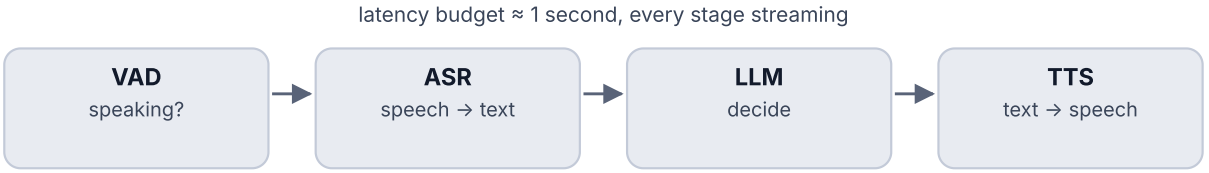
Combine cheap extraction with a VLM for the hard pages, and keep a human on uncertain fields.

5. Voice AI: the real-time pipeline

1. **VAD** (voice activity detection): is someone speaking?
2. **ASR** (speech-to-text)
3. **LLM** (with tools and RAG)
4. **TTS** (text-to-speech)

- **Speech-to-speech** models go audio-in, audio-out: lower latency and keep tone, but harder to control and evaluate
- **Latency budget ~1 second**: stream every stage (partial transcripts, first LLM tokens straight into TTS), short spoken-style answers
- Feels human: **barge-in** (the user can interrupt), good turn detection, noise and dialect handling, reading critical details back
- Transport: **WebSockets or WebRTC** (two-way audio), not SSE
- Metrics: **word error rate per dialect**, p95 response latency, task completion, transfer-to-human rate

✎ DIAGRAM



A voice agent is a latency budget, plus barge-in so callers can interrupt.

SAY IT

"For Arabic I'd budget a token multiplier, use a multilingual embedding model with normalisation, and evaluate per dialect. A voice agent is a latency budget: VAD, streaming ASR, LLM and streaming TTS over WebSockets, with barge-in."